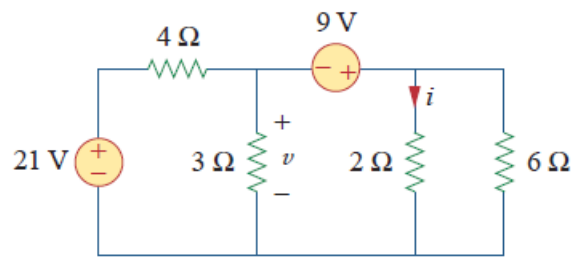


CSE251: ASSIGNMENT-1

Full Marks: 60

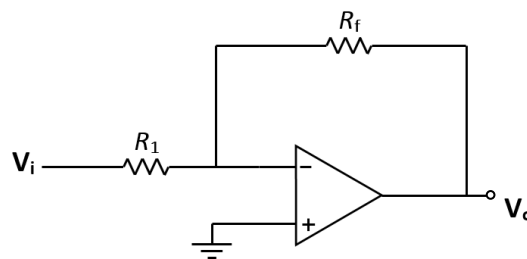
Q1. [CO2]

- Draw** the alternative representation of circuit-1 .
- Analyze** the Circuit from question-1 to determine v and i .



Circuit-1

Q2. [CO2]



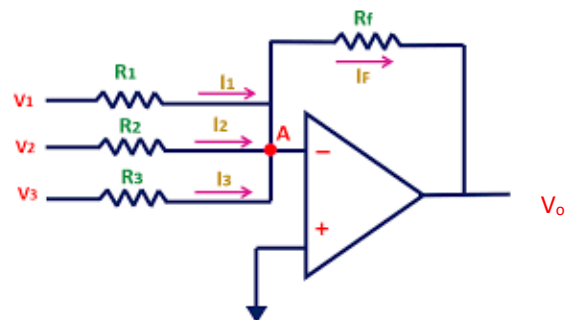
Circuit-2

For the circuit shown above , $R_1=2k$, $R_f=6k$, $+V_s= +15v$, $-V_s= -15v$

- Find** the value of V_o if $V_i=4v$
- Draw** the graph of V_o with proper labels if $V_i= 4\sin(\omega t)$
- Draw** the graph of V_o with proper labels if $V_i= 6\sin(\omega t)$

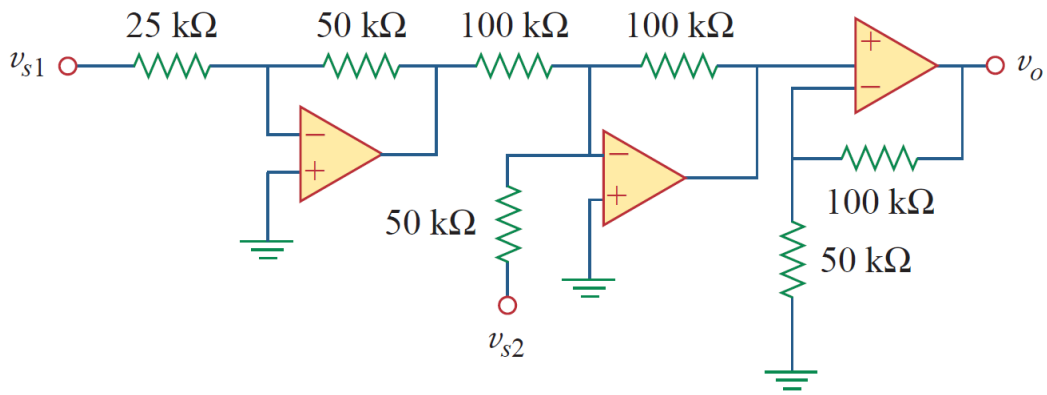
Q3. [CO2]

- Analyze** circuit-3 and derive the expression for output voltage V_o in terms of inputs.
- If $V_1= 1v$, $V_2= 2v$ and $V_3= 1.5v$ and all the resistors have same value, calculate V_o .



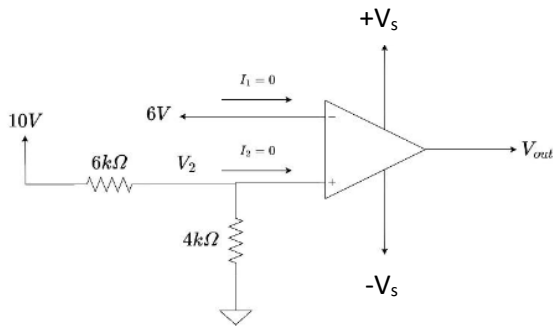
Circuit-3

Q4. [CO2]

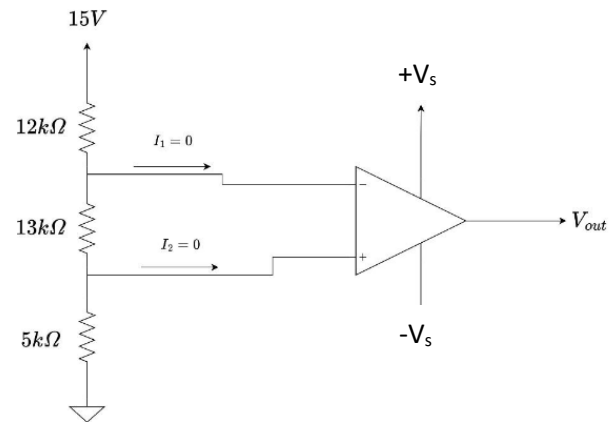


Analyze Circuit to Find V_o in terms of input V_{s1} and V_{s2} .

Q5. [CO2] Analyze the following circuit to find the output V_{out} , where $+V_s = 15\text{V}$ and $-V_s = -15\text{V}$



Circuit-a



Circuit-b

Q6. [CO3] Design a circuit using OP-Amp to implement the following functions

- $f = -5x$
- $f = 20x$
- $f = -3x - 5y - z$